PRACTICAL TITLE:
Analysis of a Local Machine Using Nmap/Zenmap

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1. AIM

To analyze a local machine using network scanning tools and identify open ports and running services.

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2. THEORY

Network scanning tools are used to discover open ports, active services, and system information on a host machine.
Nmap is a widely used tool for network discovery and security auditing.
Zenmap is the graphical interface of Nmap that helps visualize scan results.

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3. PROCEDURE

1. Open Zenmap as Administrator
2. Enter target IP address: `127.0.0.1` (local machine)
3. Select scan profile: **Quick scan**
4. Click on **Scan**
5. Observe results in the Nmap Output window

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4. OBSERVATION / OUTPUT

Starting Nmap 7.99 at 2026-04-10 10:36 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000102s latency).

Not shown: 96 closed tcp ports (reset)

```
PORT     STATE  SERVICE
135/tcp  open   msrpc
445/tcp  open   microsoft-ds
3306/tcp open   mysql
3389/tcp open   ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
```

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5. RESULT (INTERPRETATION)

- The local machine (127.0.0.1) is active and responding.
- Multiple TCP ports are open:
  - **135/tcp → MSRPC service**
  - **445/tcp → Microsoft-DS (file sharing)**
  - **3306/tcp → MySQL database service**
  - **3389/tcp → Remote Desktop Protocol**
- Total 96 ports are closed and not shown.
- The scan completed successfully in a very short time (0.11 seconds).

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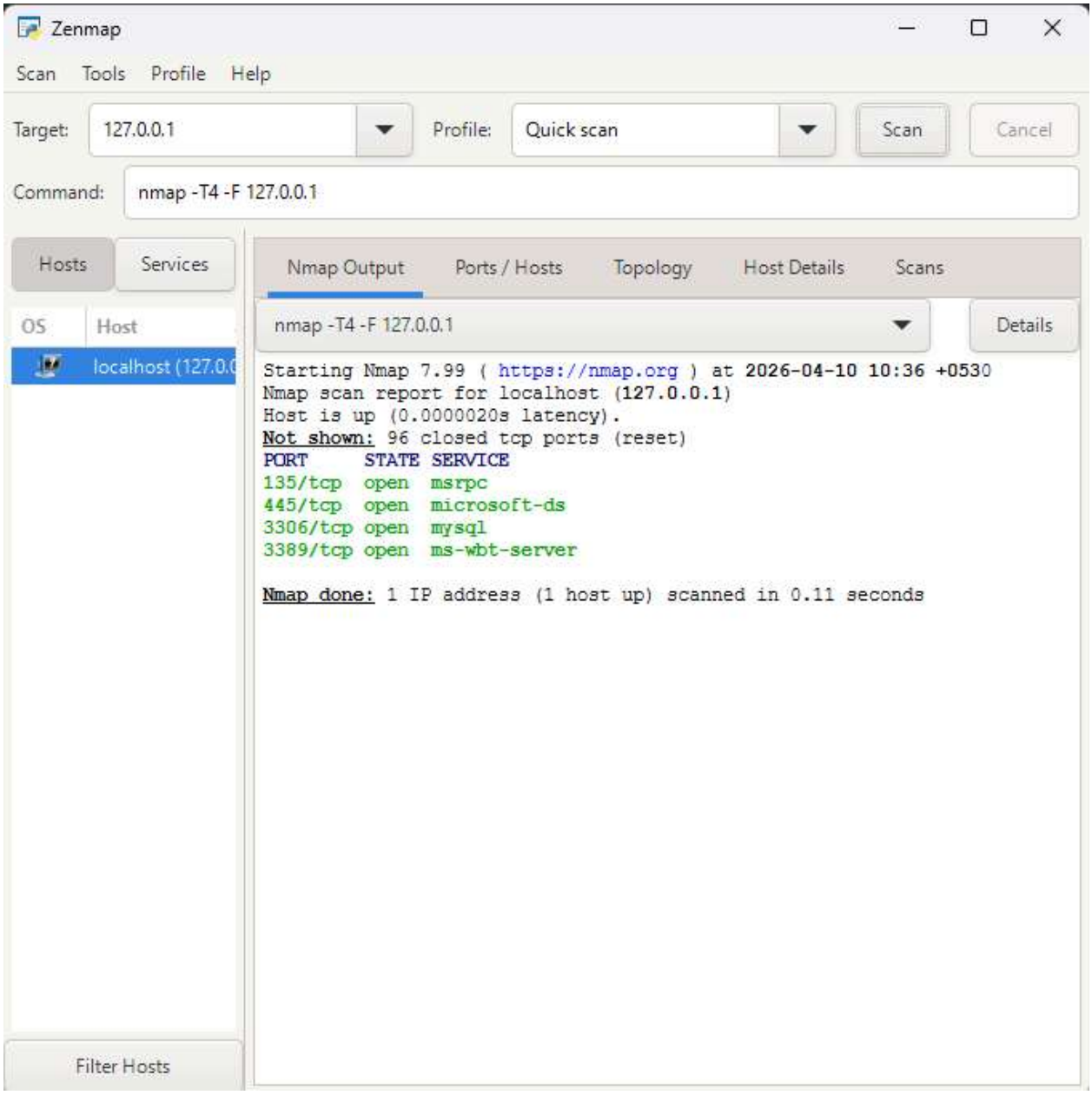
6. FINAL RESULT

The system was successfully scanned using Nmap/Zenmap, and active services along with open ports were identified on the local machine.


Zenmap
Scan Tools Profile Help
Target: 127.0.0.1
Profile: Quick scan
Scan
Cancel
Command: nmap -T4 -F 127.0.0.1
Hosts
Services
OS
Host
localhost (127.0.0
Nmap Output
Ports / Hosts
Topology
Host Details
Scans
nmap -T4 -F 127.0.0.1
Details
Starting Nmap 7.99 (https://nmap.org) at 2026-04-10 10:36 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000020s latency).
Not shown: 96 closed tcp ports (reset)
PORT STATE SERVICE
135/tcp open msrpc
445/tcp open microsoft-ds
3306/tcp open mysql
3389/tcp open ms-wbt-server
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
Filter Hosts